Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/16/2024
PI Name:	Schwertz, Hansjorg M.D., Ph.D.		
Project Title:	Megakaryocytes Orbiting in Outer Space and Near Earth: The MOON Study		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Zip Code:	84108-1263	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	12/01/2021	**End Date:**	11/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2025 per NSSC information (Ed., 12/30/24).		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Rondina, Matthew M.D. (University of Utah, Salt Lake City) Rowley, Jesse Ph.D. (University of Utah, Salt Lake City)		
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Task Description:	Megakaryocytes (MKs) and their progeny, platelets (PLTs), are dynamic effector cells with recently discovered novel functions, which bridge the inflammatory, immune, and hemostatic continuum. Changes in bone marrow MKs, resulting in low PLT numbers, (thrombocytopenia, which occurs in astronauts during spaceflight) are associated with dysregulated host inflammatory/immune responses. MKs and PLTs sense and respond to environmental cues. MKs also differentially invest developing PLTs with RNAs and proteins that alter functions of newly-released cells, influencing cellular and host responses. Surprisingly, there is a paucity of data regarding in-flight, long-term dynamics of MK development and function, as well as PLT function and production. Given previously identified and published space-travel associated risks on dysregulated inflammation, immune responses, thrombus formation, and hemostatic systems, filling this critical knowledge gap is important for the health of spaceflight crewmembers during and after missions. Moreover, as other blood cells (e.g., red blood cells, leukocytes, etc.) may be altered by microgravity, data generated are likely to contribute to our understanding of how spaceflight affects other hematopoietic processes. This proposal is based on our robust preliminary data demonstrating that conditions mimicking microgravity (rotating wall vessel culture, RWVC) markedly alter human MK morphology and gene expression. We hypothesize that microgravity will re-program MKs and newly-released PLTs, resulting in critical changes in their transcriptome, proteome, and alterations in PLT number and function. We will determine how microgravity and space radiation conditions on board the International Space Station (ISS) alter human MK and PLT maturation/production, gene expression (DNA, RNA, and protein), and cellular function. We will study in vitro human hematopoietic progenitor cell (HPC)-derived MKs in Earth-based experiments under standard or microgravity conditions. In parallel, human MKs will be studied on the ISS. Integrated, cutting-edge OMICS toolsets (e.g., RNA-sequencing and ribosomal footprinting [Ribo-seq]), comprehensive morphologic studies, and cell production kinetic studies will be used. They will provide unprecedented insight into adaptation processes needed for MK and PLT function under conditions experienced by humans performing spaceflights. These studies will directly address crew health concerns that currently limit human space exploration and will assist in developing targeted countermeasures. This proposal concurs with the major National Research Council (NRC) Decadal Survey Recommendations for cellular and molecular biology studies using state- of-the-art tools coupled with systems biology, and for studies evaluating the physiological interplay of cardiopulmonary and immune functions during application of spaceflight. Furthermore, we will address goals of the NASA Space Biology Science Plan 2016-2025, including: (1) determine the effects of the space environment on DNA function, (2) develop a systems biology-based understanding of the cellular and molecular changes to explain how gravitational changes in spaceflight effects organisms and causes phenotypic changes, and (3) identify how spaceflight affects the ability of cells to generate and maintain their complex internal cyto-architecture, processes critical for MKs and PLTs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from the proposal o have immediate mechanistic implications for the development of countermeasures for unexpected clot formation, and dysregulated systemic inflammation in ISS crew members. o may be broadly applicable to other cells, such as leukocytes, monocytes, and red blood cells, that are altered in spaceflight settings. o allowing us to translate ex vivo findings into a relevant in vivo situation via comparing longitudinal RNA expression studies in human PLTs and durable reprogramming of PLTs in astronauts (pre- vs. post-flight) mitigating cell-autonomous changes only observed in culture systems. o Identified target genes across ground and flight, cell culture and astronaut projects will be verified for functional and systems biology relevance by using cell culture CRISPR-Cas9 approaches. • Exploration Benefits o Advance knowledge will help the development of countermeasures for unexpected clot formation, and dysregulated systemic inflammation in spaceflight crew members. o Obtained data will be translatable to address known “ground-based” dysfunctions of human MKs and PLTs, may lead to the discovery of new drug targets and therapeutic interventions, and initiate new research avenues. • Earth Benefits o Since results generated may be broadly translatable to other cell types, such as leukocytes, monocytes, and red blood cells, the analysis data sets will be applicable to numerous inflammatory and pro-thrombotic conditions observed in the medical arena. Furthermore, the direct comparison of ground-simulations and ISS-flight experiments, and the integration of ex vivo findings will help in validating commonly used spaceflight simulation cell culture conditions.
	The progress and accomplishments will be divided into (1) Administrative/Organizational, and (2) Experimental tasks. (1) ADMINISTRATIVE/ORGANIZATIONAL ACCOMPLISHMENTS o Crew consent forms were signed, and blood draw questionnaires were collected. o Experimental Research Design (ERD) was finalized and signed on 01/09/2024. o Science Verification Test (SVT) was concluded, and the report submitted on 02/09/2024. o Science Verification Test Readiness Review (SVTRR/PFT) was approved on 03/05/2024. o Science Requirements Document (SRD) was revised and revision signed and accepted on 03/14/2024. o Experiment Verification Test Readiness Review (EVTRR) was approved on 03/29/2024. o Experiment Verification Test (EVT) was planned according to proposed protocol and was performed from 04/03 – 04/04/24 and 04/12 – 04/23/24 o Science Readiness for Flight Review (SRFR) took place on 05/16/24, and received approval by BPS. o NASA Kennedy Space Center (KSC) Science Lab PSRD was filed on 05/24/2024. o Payload phase started and science communication was sent to NASA on 06/21/24. o NASA Johnson Space Center (JSC) Institutional Review Board (IRB) of record was extended until 06/2025. o NASA Space Radiation Laboratory (NSRL) rerun was re-re-scheduled from Fall 2024, since it would have collided with launch of experiment to the International Space Station (ISS), to the Spring 2025 campaign. o Microscope test at Leidos was organized, supplies were shipped, and the test was performed on 07/10/24. o Supplies for on-orbit experiment are ordered per 08/2024. o Launch of experiment was scheduled. o All team members filed their forms to be badged at KSC. o One consented human research subject was removed from flight, and we are awaiting a new assignment, new baseline blood draw, or assignment of a different human research subject for future crews. (2) EXPERIMENTAL ACCOMPLISHMENTS: The following experiments were accomplished and generated valuable insights and data. Experiments related to Specific Aim 1: o Experiments using n=10 mobilized adult CD34+ hematopoietic stem cell cultures were performed, comparing standard suspension versus rotating wall vessel culture. We were able to demonstrate an appropriate increase in cell numbers, and stable cell viability. o Flowcytometric data were analyzed, as well as cell morphology and ability to form proplatelet extension by means of live-microscopy and confocal immuno-cytochemistry. o After the 11-day culture period,

Task Progress:	proplatelet formation was analyzed and demonstrated a significantly increased percentage of proplatelet forming cells in the rotating culture when compared to cells cultured under standard conditions. o Data gathered from the total RNA sequencing runs were analyzed, and verification experiments targeting transcripts of interest were performed, including functional assays and Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR-Cas9) approaches. o The rerun of the simplified Galactic Cosmic Ray simulation (simGCRsim) experiment (due to anomaly experienced during the initial run) was applied for and will take place in Spring of 2025, to avoid collision with the on-orbit experiment. Experiments related to Specific Aim 2: Several standard ground cell culture conditions need to be adjusted due to in-flight conditions on board the ISS. o All proposed SVT experiments were finished and reported. o EVT was performed from 04/03-04/04/24 and 04/12-04/23/24. First, live microscopy images were collected on day 7. Imaging was performed according to the operational protocol, using microscope similar to the on-orbit microscope with 20x objective, and custom stage holder. Imaging resulted in appropriate detection of cells, demonstration of proplatelet formation, and assessment of good cell health. o On the last day of culture, cytoskeletal live staining was performed and proplatelet extension could be clearly identified. o Viabilities and cell numbers for all EVT cell cultures were assessed. Viability was comparable with SVT results, and within the expected range. o The RNA Integrity Number (RIN) were analyzed for all samples. All measured RIN were within the acceptable range as defined by the mission success criteria. o Fixation in paraformaldehyde (PFA), storage of CD34+ cells at 4°C, and subsequent layering onto slides resulted in preserved cell architecture, tubulin staining, and demonstration of proplatelet extension. Experiments related to Specific Aim 3: o Baseline blood draws (L-[60-90]) for four crew members were performed. o Human research subject platelets were isolated and baseline aggregometry and flow cytometry to assess surface expression of hallmark receptors were performed. o Samples for RNA and ribosome-protected fragment (RPF) sequencing were prepared and stored. Such analyses will be performed once all samples of this Aim are collected to reduce sequencing bias. o Scanning and transmission electron microscopy, as well as confocal microscopy, were performed with expected imaging results demonstrating baseline platelet morphology and ultrastructure.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Articles in Peer-reviewed Journals	Seylani A, Galsinh AS, Tasoula A, RI A, Camera A, Calleja-Agius J, Borg J, Goel C, Kim J, Clark KB, Das S, Arif S, Boerrigter M, Coffey C, Szewczyk N, Mason CE, Manoli M, Karouia F, Schwertz H, Beheshti A, Tulodziecki D. "Ethical considerations for the age of non-governmental space exploration." Nat Commun. 2024 Jun 11;15:4774. Review. https://doi.org/10.1038/s41467-023-44357-x ; PMID: 38862473; PMCID: PMC11166968 , Jun-2024